GATE- ALL BRANCHES

ENGINEERING MATHEMATICS

Linear Algebra

Lecture No. 5

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Recap of previous lecture

→ Rank

→ linear combination of Vectors → $\sum_{i=1}^{n} c_i \vec{x}_i = \vec{X}$.

- → linearly Dependent → $\vec{x}_j = \sum_{\substack{i=1 \\ i \neq j}}^{n} c_i \vec{x}_i$.
- → Linearly Independent → $\sum_{i=1}^{n} c_i \vec{x}_i = \vec{0}$ only for $c_1 = c_2 = \dots = c_n = 0$.

For $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n \in \mathbb{R}^3$;

then consider $|\vec{x}_1 \; \vec{x}_2 \; \vec{x}_3|$;

if $|\vec{x}_1 \; \vec{x}_2 \; \vec{x}_3| = 0$; then Vectors are linearly dependent.

Topics to be Covered

Topic Vector Spaces

Topic System of Linear Equations

$\rightarrow \quad A = \begin{bmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 1 & 4 \\ 3 & 1 & 1 & 2 \end{bmatrix}$

$$\rightarrow \det(A) = 2\cdot(-1)^{1+1}\cdot\begin{vmatrix} 4 & 3 & 2 \\ 1 & 1 & 4 \\ 1 & 1 & 2 \end{vmatrix} + 3\cdot(-1)^{1+2}\cdot\begin{vmatrix} 1 & 3 & 2 \\ 2 & 1 & 4 \\ 3 & 1 & 2 \end{vmatrix}$$

$$+ 1\cdot(-1)^{1+3}\cdot\begin{vmatrix} 1 & 4 & 2 \\ 2 & 1 & 4 \\ 3 & 1 & 2 \end{vmatrix} + 4\cdot(-1)^{1+4}\cdot\begin{vmatrix} 1 & 4 & 3 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{vmatrix}$$

→ linear Algebra & Applications → Gilbert M strang

→ linear Algebra & optimization → Charu. Agarwal

Vector spaces

A space 'V' of vectors is said to be a Vector Space if the space closes under Addition & Scalar Multiplication.

$$V = \{ \vec{v}_1, \vec{v}_2, \ldots \}$$

Where For $\vec{v}_1, \vec{v}_2 \in V$; $\vec{v}_1 + \vec{v}_2 \in V$. → closure under Addition.

For a scalar 'α' & $\vec{v} \in V$; $\alpha\vec{v} \in V$ → closure under Multiplication.

→ If $\alpha = 0$; $\Rightarrow \vec{0} \in V$; If $\alpha = -1$; $\Rightarrow -\vec{v} \in V$.

→ If $\vec{v}_1, \vec{v}_2 \in V \Rightarrow \alpha_1\vec{v}_1, \alpha_2\vec{v}_2 \in V$

If $\vec{v}_1, \vec{v}_2 \in V$ then $\alpha_1\vec{v}_1, \alpha_2\vec{v}_2 \in V$

$$\Rightarrow \alpha_1\vec{v}_1 + \alpha_2\vec{v}_2 \in V.$$

If $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n \in V$ then any linear combination

$$\sum_{i=1}^{n} \alpha_i \vec{v}_i \in V.$$

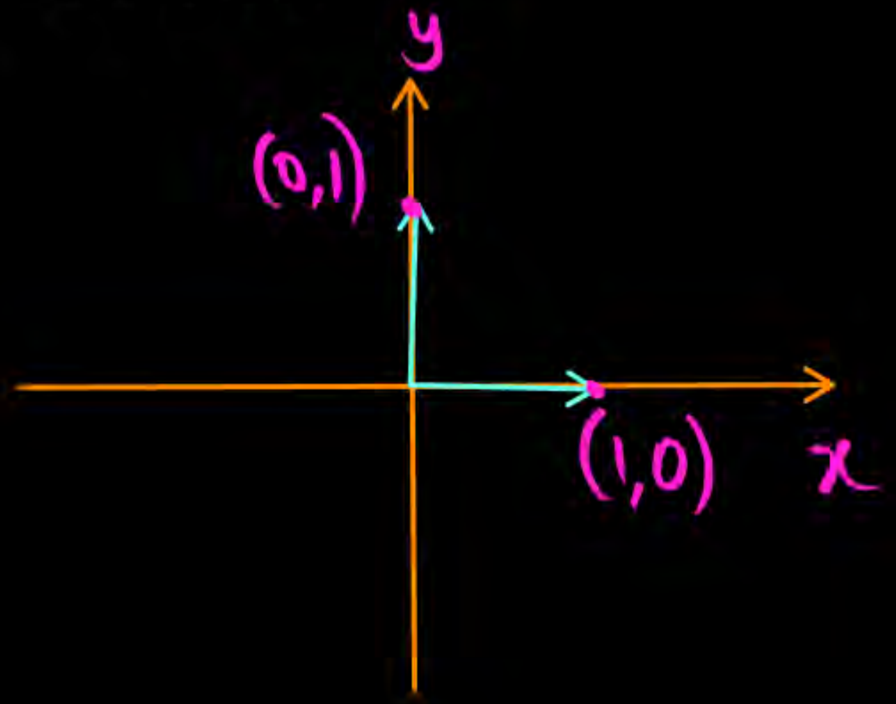
→ **Subspace of 'V':**

If 'V' is a Vector space, then a Sub Vector space formed by Picking Some Vectors from 'V' such that this subspace is also closed under addition & Scalar Multiplication.

$$V = \text{span}\left\{ \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix} \right\}$$

$$\Rightarrow V = \left\{ \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \alpha_1\begin{bmatrix}1\\0\end{bmatrix} + \alpha_2\begin{bmatrix}0\\1\end{bmatrix} \right\}$$

$\Rightarrow \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}$ spans complete 2-D space.

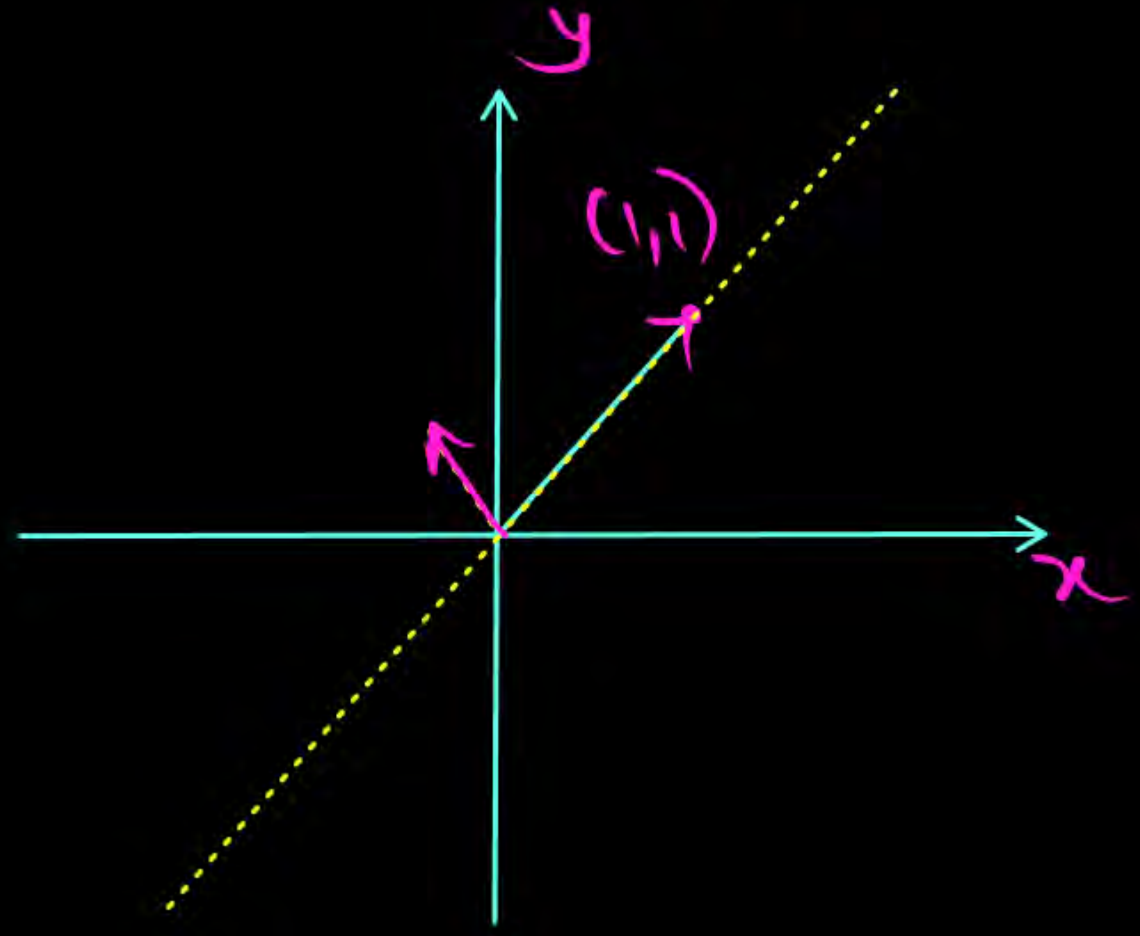


$\rightarrow$ If $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$ are 'n' vectors, then $\text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is the space that contains all the linear combinations of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$.

Ex: Say $\vec{V}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\vec{V}_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

then Any linear combination;

$$\vec{V} = \underbrace{(\alpha_1 + 2\alpha_2)}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



#Q. Select all the sets which represents a vector subspace?

(A) $V = \left\{ \begin{bmatrix} 2x \\ y+z \\ 3z \end{bmatrix}; x, y, z \in R \right\}$

(B) $V = \left\{ \begin{bmatrix} t \\ s \\ \log s \end{bmatrix}; t, s \in R \right\}$

(C) $V = \left\{ \begin{bmatrix} 1 \\ p \\ q \end{bmatrix}; p, q \in R \right\}$

(D) $V = \left\{ \begin{bmatrix} 0 \\ p \\ q \end{bmatrix}; p, q \in R \right\}$

Sol:

A) $V = \left\{ \begin{bmatrix} 2x \\ y+z \\ 3z \end{bmatrix} ; x, y, z \in \mathbb{R} \right\}$.

consider $\vec{v}_1 = \begin{bmatrix} 2x_1 \\ y_1+z_1 \\ 3z_1 \end{bmatrix}$; $\vec{v}_2 = \begin{bmatrix} 2x_2 \\ y_2+z_2 \\ 3z_2 \end{bmatrix}$

$\Rightarrow \vec{v}_1 + \vec{v}_2 = \begin{bmatrix} 2(x_1+x_2) \\ (y_1+y_2)+(z_1+z_2) \\ 3(z_1+z_2) \end{bmatrix} \in V$,

$\Rightarrow$ 'V' is a subspace.

B) $V = \left\{ \begin{bmatrix} t \\ s \\ \log s \end{bmatrix} ; t, s \in \mathbb{R} \right\}$.

consider $\vec{v}_1 = \begin{bmatrix} t_1 \\ s_1 \\ \log s_1 \end{bmatrix}$; $\vec{v}_2 = \begin{bmatrix} t_2 \\ s_2 \\ \log s_2 \end{bmatrix}$

$\Rightarrow \vec{v}_1 + \vec{v}_2 = \begin{bmatrix} t_1+t_2 \\ s_1+s_2 \\ \boxed{\log s_1 + \log s_2} \end{bmatrix}$

$\because \log s_1 + \log s_2 \neq \log(s_1+s_2)$

$\Rightarrow$ 'V' is not a subspace.

c) $V = \left\{ \begin{bmatrix} 1 \\ p \\ q \end{bmatrix} ; \ p, q \in \mathbb{R} \right\}$

Consider $\vec{V}_1 = \begin{bmatrix} 1 \\ p_1 \\ q_1 \end{bmatrix}$; $\vec{V}_2 = \begin{bmatrix} 1 \\ p_2 \\ q_2 \end{bmatrix}$

$\Rightarrow \vec{V}_1 + \vec{V}_2 = \begin{bmatrix} 2 \\ \underline{p_1 + p_2} \\ \underline{q_1 + q_2} \end{bmatrix}$ $\to \neq 1$

$\Rightarrow$ This is Not a subspace.

D) $V = \left\{ \begin{bmatrix} 0 \\ p \\ q \end{bmatrix} ; \ p, q \in \mathbb{R} \right\}$.

consider $\vec{V}_1 = \begin{bmatrix} 0 \\ p_1 \\ q_1 \end{bmatrix}$; $\vec{V}_2 = \begin{bmatrix} 0 \\ p_2 \\ q_2 \end{bmatrix}$

$\Rightarrow \vec{V}_1 + \vec{V}_2 = \begin{bmatrix} 0 \\ (p_1 + p_2) \\ (q_1 + q_2) \end{bmatrix} \in V$

$\Rightarrow$ 'V' is a Sub-space.

→ Basis of a Vector space 'V':

For the Vector space 'V' containing $\vec{v}_1, \vec{v}_2, \vec{v}_3, \ldots, \vec{v}_n, \ldots$

the set $\{\underbrace{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_r}\}$ is said to be basis of the space 'V' if

a) $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_r$ are the Maximum Possible linearly independent vectors in 'V'.

i.e If $\vec{v}_{r+1}$ is included then $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_r, \vec{v}_{r+1}$ are linearly dependent.

b) $\text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_r\} = V$

Ex: $V = \text{span}\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}\right\}$

Consider $c_1\begin{bmatrix}1\\0\\0\end{bmatrix} + c_2\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}c_1\\c_2\\0\end{bmatrix} = \vec{0}$.

$\Rightarrow c_1 = 0;\ c_2 = 0$

clearly $\begin{bmatrix}1\\0\\0\end{bmatrix}; \begin{bmatrix}0\\1\\0\end{bmatrix}$ are linearly Independent.

But $\text{span}\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}\right\} \neq \mathbb{R}^3$.

$\vec{x} = \begin{bmatrix}a\\b\\c\end{bmatrix}$

$\Rightarrow \left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}\right\}$ is NOT a basis for $\mathbb{R}^3$.

Ex: consider $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$.

$V = \mathbb{R}^3$.

Consider $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$; $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$; $\vec{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\cancel{\vec{x}}^{\,0} = \underset{\sim}{c_1}\vec{x}_1 + \underset{\sim}{c_2}\vec{x}_2 + \underset{\sim}{c_3}\vec{x}_3 + \underset{\sim}{c_4}\vec{x}_4$

Not all zero simultaneously

Consider $\left| \vec{v}_1 \;\; \vec{v}_2 \;\; \vec{v}_3 \right| = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \neq 0$

$\Rightarrow$ Given Vectors are linearly Independent.

Consider $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \boxed{\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}} \Rightarrow \cancel{\alpha_1}^{\,2} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \cancel{\alpha_2}^{\,-1} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \cancel{\alpha_3}^{\,0} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \cancel{\alpha_4}^{\,-1} \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = \vec{0}$

This vector makes complete combination L. Dependent.

Also $\text{span}\left\{ \underbrace{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}}_{\text{Basis of } \mathbb{R}^3} \right\} = \mathbb{R}^3$.

Note: The Basis for a Vector space 'V' is NOT unique.

Ex: For $\mathbb{R}^{3}$, the Basis can be $\left\{ \begin{bmatrix}-1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\-1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\-1\end{bmatrix} \right\}$,

$\left\{ \begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\-1\end{bmatrix} \right\}$, $\left\{ \begin{bmatrix}2\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\-1\end{bmatrix} \right\}$.

In any basis, No. of elements is Same. (Dimension of Vector space)

Dimension of a Vector Space:

If 'V' is a Vector Space spanned by the basis $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_r\}$, then dimension of $V = r$.

It is denoted as $\boxed{\dim(V) = r.}$

i.e The Number of linearly independent vectors that span the full Vector Space 'V' is it's dimension.

→ For 'm' Vectors in $n-D$ Space; $A_{m\times n} = [\vec{v}_1^T \vec{v}_2^T \ldots \vec{v}_n^T]_{m\times n}$.

$\Rightarrow \rho(A_{m\times n}) = r$.

System of linear equations

Consider a System of linear Equations;

$$\left.\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\
a_{31}x_1 + a_{32}x_2 + \cdots + a_{3n}x_n &= b_3 \\
\vdots \qquad\quad \vdots \qquad\qquad\quad \vdots \quad &\quad\ \vdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m
\end{aligned}\right\} \longrightarrow$$

N-unknowns.

&

'm' Equations.

$$\Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \Rightarrow A_{m\times n}\,\vec{x}_{n\times 1} = \vec{b}_{m\times 1}$$

Coefficient Matrix

Variable Matrix

Constant Matrix.

A linear system of equations $A\vec{x} = \vec{b}$; is said to be Consistent if the system has a Solution. (May be unique (or) Infinitely Many).

Conditions for Consistency:

① If $\rho(A) = \rho(A|\vec{b}) = n$; then the system has unique Solution.

② If $\rho(A) = \rho(A|\vec{b}) < n$; then the System $A\vec{x} = \vec{b}$ has infinitely Many solutions.

③ If $\rho(A) \neq \rho(A|\vec{b})$; then the System $A\vec{x} = \vec{b}$ has No solution.

$\underbrace{[A|\vec{b}]}_{m\times(n+1)} \rightarrow$ Augmented Matrix.

$$[A|\vec{b}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ a_{31} & a_{32} & \cdots & a_{3n} & b_3 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

Ex: Consider the Equation.

$$2x - y = 3 \rightarrow (1)$$
$$x + 2y = 4 \rightarrow (2)$$

$$\Rightarrow \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \Rightarrow A\vec{x} = \vec{b}.$$

$$\rho(A) = 2 \; (\because \det(A) \neq 0)$$

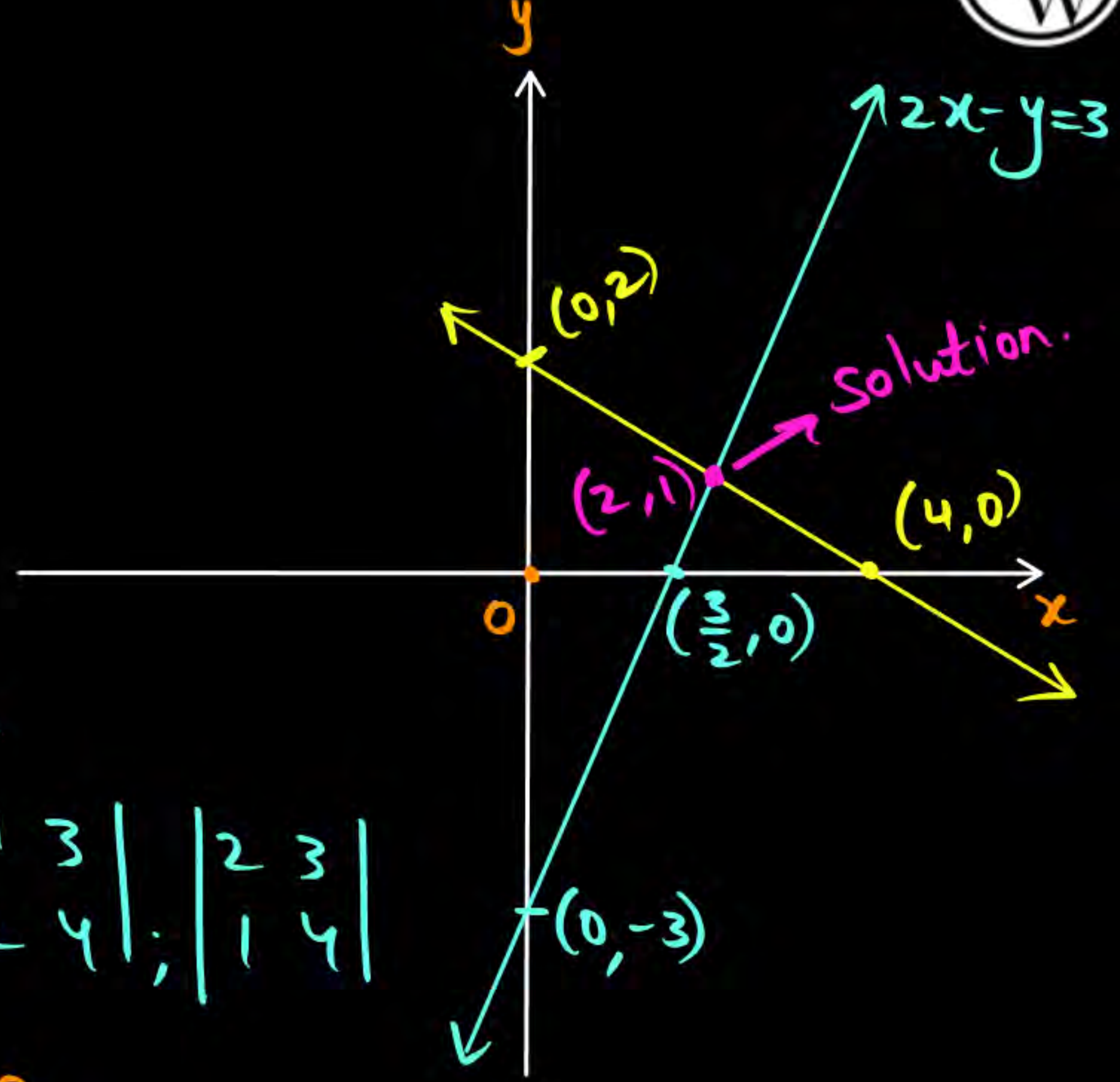
Consider $[A|\vec{b}] = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & 4 \end{bmatrix}_{2\times 3}$ $\Rightarrow$ 2x2 Minors.

$$\boxed{\begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix}}, \begin{vmatrix} -1 & 3 \\ 2 & 4 \end{vmatrix}; \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix}$$

$\det(A) \neq 0$

$$\Rightarrow \boxed{\rho(A) = \rho(A|\vec{b}) = 2}$$

$\Rightarrow$ The system $A\vec{x} = \vec{b}$ has unique Solution.



Note:

→ If System has 'n' equation with 'n' unknowns (Say $A_{n\times n}\vec{x}_{n\times 1} = \vec{b}_{n\times 1}$).

and $\det(A) \neq 0$; $\Rightarrow \underbrace{\rho(A) = \rho(A|\vec{b}) = n}$

⇓

System has unique solution.

→ For $A_{n\times n}\vec{x}_{n\times 1} = \vec{b}_{n\times 1}$ with $\det(A) \neq 0$; the system always has a unique solution.

$\underbrace{[A|\vec{b}]_{n\times(n+1)}}$

↓

Max. Possible Minors are of order $n\times n$

↓

out of $n\times n$ minors, 1 Minor is definitely $\det(A)$

For $\det(A) \neq 0$

$\Rightarrow \rho(A|\vec{b}) = n$.

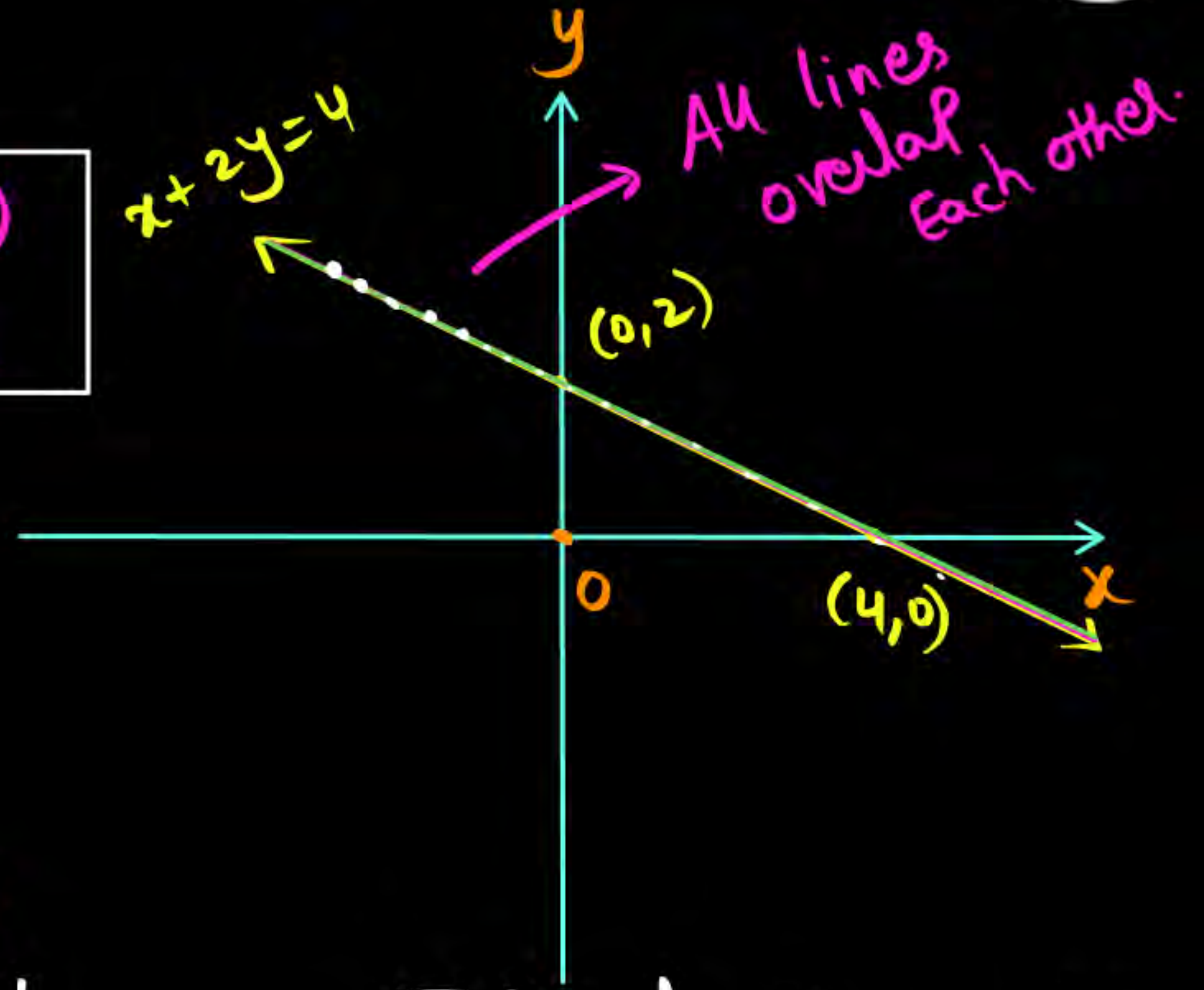
Ex:

Consider.

$$\left.\begin{aligned} x+2y &= 4 \\ 2x+4y &= 8 \\ 3x+6y &= 12 \end{aligned}\right\} \rightarrow \text{3 Eqns with 2 unknowns.}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}_{3\times 2} \begin{bmatrix} x \\ y \end{bmatrix}_{2\times 1} = \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix}_{3\times 1} \Rightarrow A\vec{x} = \vec{b}$$

$$\boxed{\begin{aligned} \Rightarrow \rho(A) &= \rho(A|\vec{b}) \\ &= 1 < 2 \end{aligned}}$$



$\rho(A)$: 2×2 Minors.

$$\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} \rightarrow 0 ; \quad \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} \rightarrow 0 ; \quad \begin{vmatrix} 2 & 4 \\ 3 & 6 \end{vmatrix} \rightarrow 0$$

$$\Rightarrow \rho(A) = 1$$

Infinitely Many solutions

$\rho(A|\vec{b})$: $[A|\vec{b}] = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 3 & 6 & 12 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} [A|\vec{b}] \sim \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \boxed{\rho(A|\vec{b}) = 1}$

Home work.

(1) check the consistency for

$$2x+4y=7$$
$$x+2y=8$$

(2) check the consistency for

$$x+2y=4$$
$$3x-y=7$$

2 mins Summary

→ Vector spaces

→ basis

→ span

→ Dimension of 'V'

→ System of Equations.
(Linear Equations).

THANK - YOU